

UNIX and Linux System Administration Handbook

One-Week DevOps / SRE Reading Cut

Goal: extract only the handbook material that directly improves infra interviews, operational judgment, and troubleshooting instincts. This is not a full-book read. It is a sharp cut for platform / DevOps / SRE roles.

What to read

- **Chapter 3 — Access Control and Rootly Powers** — Users, groups, sudo, privilege boundaries, PAM/ACL awareness, safe root practices.
- **Chapter 4 — Process Control** — Processes, signals, states, jobs, daemons, /proc, ps/top, strace/lsof basics.
- **Chapter 5 — The Filesystem** — Inodes, links, mounts, file metadata, permissions, space, practical filesystem behavior.
- **Chapter 10 — Logging** — Syslog/journald mental model, log routing, useful log locations, debugging habits.
- **Chapter 13 — TCP/IP Networking** — Addressing, routing, interfaces, DNS path intuition, troubleshooting flow, NAT/firewall basics.
- **Chapter 16 — DNS** — Zones, records, recursion vs authority, caching, failure modes, dig/nslookup workflow.
- **Chapter 20 — Storage** — Disks, partitions, LVM/RAID/filesystem choices, backups, reliability tradeoffs.
- **Chapter 27 — Security** — SSH, authentication hygiene, firewalls, surface reduction, operational hardening basics.
- **Chapter 28 — Monitoring** — What to measure, host vs service monitoring, alert signal quality, monitoring architecture.
- **Chapter 29 — Performance Analysis** — Bottleneck thinking, CPU/memory/disk/network diagnosis, slow-server workflow.

Skip for now

- Boot/install details unless you feel weak on systemd and service startup.
- Email, printing, LDAP, SMB, web hosting internals, and organization/management chapters.
- Deep distro-specific package-management rabbit holes beyond what you already use at work.
- Any section that turns into a tool catalog instead of a mental model.

Required subtopics

Chapter	Required subtopics	Why it matters
Ch. 3	users, groups, sudo, ACLs, PAM, privilege escalation boundaries	Explains access failures, least privilege, and safe production access.
Ch. 4	process states, signals, kill/killall, ps/top, nice, /proc, lsof, strace	Core incident response and “what is this process doing?” skills.
Ch. 5	inodes, hard vs soft links, mount points, df/du, metadata, permissions	Makes filesystem behavior feel mechanical instead of magical.
Ch. 10	syslog/journald basics, log routing, rotation, key log files	Turns debugging from random grep panic into a workflow.
Ch. 13	IP addressing, routing tables, interfaces, ARP-level intuition, NAT, packet path	Needed for load balancers, DNS weirdness, and network triage.
Ch. 16	authoritative vs recursive DNS, records, TTLs, caching, dig output	High-yield for SRE, platform, service discovery, and outages.
Ch. 20	RAID/LVM/filesystems, mount strategy, capacity, snapshots/backups	Storage tradeoffs show up constantly in ops and design talks.
Ch. 27	SSH, host hardening, user hygiene, firewall basics, secrets discipline	Enough security fluency to not sound naive in infra rounds.
Ch. 28	host metrics, service checks, alerting philosophy, monitoring scope	Connects directly to observability and incident prevention.
Ch. 29	CPU vs memory vs I/O vs network bottlenecks, perf workflow	The chapter that makes you sound like an operator, not a reader.

One-week reading order

1. **Day 1 — Chapter 3 + Chapter 4:** Access, privilege, users/groups, sudo, signals, process states, ps/top/lsof/strace.
2. **Day 2 — Chapter 5:** Filesystem model, links, mounts, permissions, space, “why is disk full?” intuition.

3. **Day 3 — Chapter 13 (first half):** Interfaces, addresses, routes, packet path, NAT, basic firewall intuition.
4. **Day 4 — Chapter 16 + selected Chapter 10:** DNS deeply; then only the logging pieces that help real troubleshooting.
5. **Day 5 — Chapter 20:** Disks, LVM/RAID/filesystems, backup/snapshot tradeoffs.
6. **Day 6 — Chapter 27 + Chapter 28:** Security basics, SSH/firewalls/account hygiene, then monitoring signal design.
7. **Day 7 — Chapter 29:** Performance analysis end-to-end. Revisit weak notes from Days 1–6.

What to extract while reading

- One command workflow per topic: e.g. process triage, DNS triage, disk triage, memory triage.
- One failure mode per chapter: stale DNS, inode exhaustion, runaway process, log flood, disk saturation.
- One interview sentence per chapter: how you would explain the concept out loud in a practical way.
- One connection to your work: Kafka/Pulsar, k8s nodes, Route 53, EKS, Terraform, on-call incidents.

Final cut

Must read	Nice only if time remains
3, 4, 5, 13, 16, 20, 27, 28, 29	10 2 if you feel weak on boot/systemd

Built for fast, high-yield prep. Read for operational judgment, not completeness.